

Finally, the sixth column shows the Poisson results for the whole sample (including zero-trade pairs).

The first point to notice is that PPML-estimated coefficients are remarkably similar using both the whole sample and the positive-trade subsample.²² However, most coefficients differ – oftentimes significantly – from those obtained using OLS. This suggests that in this case, heteroskedasticity (rather than truncation) is responsible for the differences between PPML results and those of OLS using only the observations with positive trade. Further evidence on the importance of the heteroskedasticity is provided by the two-degrees-of-freedom special case of White’s test for heteroskedasticity (see Wooldridge, 2002, p. 127), which leads to a test statistic of 476.6 and to a p-value of zero. That is, the null hypothesis of homoskedastic errors is unequivocally rejected.

Poisson estimates reveal that the coefficients on importer’s and exporter’s GDPs in the traditional equation are not, as generally believed, close to 1. The estimated GDP elasticities are just above 0.7 (s.e. = 0.03). OLS generates significantly larger estimates, especially on exporter’s GDP (0.94, s.e. = 0.01). Although all these results are conditional on the particular specification used,²³ it is worth pointing out that unit-income elasticities in the simple gravity framework are at odds with the observation that the trade-to-GDP ratio decreases with total GDP, or, in other words, that smaller countries tend to be more open to international trade.²⁴

The role of geographical distance as trade deterrent is significantly larger under OLS; the estimated elasticity is -1.17 (s.e. = 0.03), whereas the Poisson estimate is -0.78 (s.e. = 0.06). This lower estimate suggests a smaller role for transport costs in the determination of trade patterns. Furthermore, Poisson estimates indicate that, after controlling for bilateral distance, sharing a border does not influence trade flows, while

²²The reason why truncation has little effect in this case is that observations with zero trade correspond to pairs for which the estimated value of trade is close to zero. Therefore, the corresponding residuals are also close to zero and their elimination from the sample has little effect.

²³This result holds when one looks at the subsample of OECD countries. It is also robust to the exclusion of GDP per capita from the regressions.

²⁴Note also that PPML predicts almost equal coefficients for the GDPs of exporters and importers.

OLS, instead, generates a substantial effect: It predicts that trade between two contiguous countries is 37 percent larger than trade between countries that do not share a border.²⁵

We control for remoteness to account for the hypothesis that larger distances to all other countries might increase bilateral trade between two countries.²⁶ Poisson regressions support this hypothesis, whereas OLS estimates suggest that only exporter’s remoteness increases bilateral flows between two given countries. Access to water appears to be important for trade flows, according to Poisson regressions; the negative coefficients on the land-locked dummies can be interpreted as an indication that ocean transportation is significantly cheaper. In contrast, OLS results suggest that whether or not the exporter is landlocked does not influence trade flows, whereas a landlocked importer experiences lower trade. (These asymmetries in the effects of remoteness and access to water for importers and exporters are hard to interpret.) We also explore the role of colonial heritage, obtaining, as before, significant discrepancies: Poisson indicates that colonial ties play no role in determining trade flows, once a dummy variable for common language is introduced. OLS regressions, instead, generate a sizeable effect (countries with a common colonial past trade almost 45 percent more than other pairs). Language is statistically and economically significant under both estimation procedures.

Strikingly, in the traditional gravity equation, preferential-trade agreements play a much smaller — although still substantial — role according to Poisson regressions. OLS estimates suggest that preferential-trade agreements rise expected bilateral trade by 63 percent, whereas Poisson estimates indicate an average enhancement effect below 20 percent.

Preferential-trade agreements might also cause trade diversion; if this is the case, the coefficient on the trade-agreement dummy will not reflect the net effect of trade agreements. To account for the possibility of diversion, we include an additional dummy, “openness,” similar to that used by Frankel (1997). This dummy takes the value 1 whenever one (or

²⁵The formula to compute this effect is $(e^{b_i} - 1) \times 100\%$, where b_i is the estimated coefficient.

²⁶To illustrate the role of remoteness, consider two pairs of countries, (i, j) and (k, l) , and assume that the distance between the countries in each pair is the same $D_{ij} = D_{kl}$, however, i and j are closer to other countries. In this case, the most remote countries, k and l , will tend to trade more between each other because they do not have alternative trading partners. See Deardoff (1998).

both) of the countries in the pair is part of a preferential-trade agreement and, thus, it captures the extent of trade between members and non-members of a preferential-trade agreement. The sum of the coefficients on the trade agreement and the openness dummies gives the net creation effect of trade agreements. OLS suggests that there is trade destruction coming from trade agreements. Still, the net creation effect is around 40 percent. In contrast, Poisson regressions provide no significant evidence of trade diversion, although the point estimates are of the same order of magnitude under both methods.

Hence, even when accounting for trade diversion effects, on average, the Poisson method estimates a smaller effect of preferential-trade agreements on trade, approximately half of that indicated by OLS. The contrast in estimates suggests that the biases generated by standard regressions can be substantial, leading to misleading inferences and, perhaps, erroneous policy decisions.²⁷

We now turn briefly to the results of the other estimation methods. OLS on $\ln(1 + T_{ij})$ and Tobit give very close estimates for most coefficients. Like OLS, they yield large estimates for the elasticity of bilateral trade with respect to distance. However, unlike OLS, they produce insignificant coefficients for the contiguity dummy. They both generate extremely large and statistically significant coefficients for the trade-agreement dummy. The first method predicts that trade between two countries that signed a trade agreement is on average 266 percent larger than that between countries without an agreement. The second predicts that trade between countries in free-trade agreements is on average 100 percent larger. NLS tends to generate somewhat different estimates. The elasticity of trade with respect to the exporter's GDP is significantly smaller than OLS, but the corresponding elasticity with respect to importer's GDP is significantly larger than OLS. The estimated distance elasticity is smaller than OLS and bigger than Poisson. As the other methods, NLS predicts a significant and large effect for free-trade agreements.

²⁷It is interesting to remark that there is a pattern in the direction of the bias generated by OLS. The bias tends to be positive for the coefficients on variables that relate to larger volumes of trade and, presumably, to larger variance. It tends to be negative for variables that deter trade and, possibly, reduce the variance.

Table 3: The Traditional Gravity Equation

Estimator	OLS	OLS	Tobit	NLS	PPML	PPML
Dependent Variable	$\ln(T_{ij})$	$\ln(1 + T_{ij})$	$\ln(a + T_{ij})$	T_{ij}	$T_{ij} > 0$	T_{ij}
Log exporter's GDP	0.938** (0.012)	1.128** (0.011)	1.058** (0.012)	0.738** (0.038)	0.721** (0.027)	0.733** (0.027)
Log importer's GDP	0.798** (0.012)	0.866** (0.012)	0.847** (0.011)	0.862** (0.041)	0.732** (0.028)	0.741** (0.027)
Log exporter's GDP per capita	0.207** (0.017)	0.277** (0.018)	0.227** (0.015)	0.396** (0.116)	0.154** (0.053)	0.157** (0.053)
Log importer's GDP per capita	0.106** (0.018)	0.217** (0.018)	0.178** (0.015)	-0.033 (0.062)	0.133** (0.044)	0.135** (0.045)
Log distance	-1.166** (0.034)	-1.151** (0.040)	-1.160** (0.034)	-0.924** (0.072)	-0.776** (0.055)	-0.784** (0.055)
Contiguity dummy	0.314* (0.127)	-0.241 (0.201)	-0.225 (0.152)	-0.081 (0.100)	0.202 (0.105)	0.193 (0.104)
Common-language dummy	0.678** (0.067)	0.742** (0.067)	0.759** (0.060)	0.689** (0.085)	0.752** (0.134)	0.746** (0.135)
Colonial-tie dummy	0.397** (0.070)	0.392** (0.070)	0.416** (0.063)	0.036 (0.125)	0.019 (0.150)	0.024 (0.150)
Landlocked-exporter dummy	-0.062 (0.062)	0.106* (0.054)	-0.038 (0.052)	-1.367** (0.202)	-0.873** (0.157)	-0.864** (0.157)
Landlocked-importer dummy	-0.665** (0.060)	-0.278** (0.055)	-0.479** (0.051)	-0.471** (0.184)	-0.704** (0.141)	-0.697** (0.141)
Exporter's remoteness	0.467** (0.079)	0.526** (0.087)	0.563** (0.068)	1.188** (0.182)	0.647** (0.135)	0.660** (0.134)
Importer's remoteness	-0.205* (0.085)	-0.109 (0.091)	-0.032 (0.073)	1.010** (0.154)	0.549** (0.120)	0.561** (0.118)
Free-trade agreement dummy	0.491** (0.097)	1.289** (0.124)	0.729** (0.103)	0.443** (0.109)	0.179* (0.090)	0.181* (0.088)
Openness	-0.170** (0.053)	0.739** (0.050)	0.310** (0.045)	0.928** (0.191)	-0.139 (0.133)	-0.107 (0.131)
Observations	9613	18360	18360	18360	9613	18360
RESET test p-values	0.000	0.000	0.204	0.000	0.941	0.331

It is noteworthy that all methods, except the PPML, lead to puzzling asymmetries in the elasticities with respect to importer and exporter characteristics (especially remoteness and access to water).

To check the adequacy of the estimated models, we performed a heteroskedasticity-robust RESET test (Ramsey, 1969). This is essentially a test for the correct specification of the conditional expectation, which is performed by checking the significance of an additional regressor constructed as $(x'b)^2$, where b denotes the vector of estimated parameters.